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CODE A

Mock JEE Main – 9 (CBT) | JEE - 2024

Date: 23/03/2024 **Maximum Marks: 300**

Timing: 9:30 AM to 12:30 PM **Duration: 3.0 Hours**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices **(A)**, **(B)**, **(C)** and **(D)**, out of which **ONLY ONE CHOICE** is correct.
4. **Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. You *will* NOT be allowed to attempt the sixth question. If you wish to attempt any other question apart from the five already attempted, then you will have to delete any one response from the five previously answered and then proceed to answer the new one.**
   The answer to each question should be **rounded off to the nearest integer**.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

Name of the Candidate (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

PART I: PHYSICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The amplitude of electric field of a plane electromagnetic wave propagating along Z-axis is $4Vm^{-1}$.

It varies with time. The average energy density of the magnetic field in Jm^{-3} is:

$(\varepsilon_0 = 8.854\times10^{-12}C^2N^{-1}m^{-2})$

(A) 25.4×10^{-12} **(B)** 30.5×10^{-12}

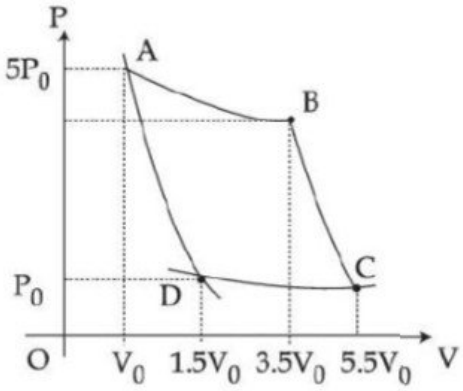
(C) 45.4×10^{-12} **(D)** 35.4×10^{-12}

2. In thermal equilibrium, the average velocity of gas molecules is:

(A) Proportional to $\sqrt{T}$ **(B)** Proportional to T^2

(C) Proportional to T^3 **(D)** Zero

3. In the reported figure, there is a cyclic process $ABCDA$ on a sample of 1 mole of a diatomic gas. The temperature of the gas during the process $A \to B$ and $C \to D$ are T_1 and T_2 $(T_1 > T_2)$ respectively.



Choose the correct option out of the following for work done if processes BC and DA are adiabatic.

(A) $W_{AB} = W_{DC}$ **(B)** $W_{AD} = W_{BC}$ **(C)** $W_{AB} < W_{CD}$ **(D)** $W_{BC} + W_{DA} > 0$

4. When light rays are incident on a prism at an angle of 45°, the minimum deviation is obtained.

If refractive index of the material of prism is $\sqrt{2}$, then the angle of prism will be:

(A) 30° **(B)** 40° **(C)** 50° **(D)** 60°

5. An organ pipe closed at one end has fundamental frequency of 1500 Hz. The maximum number of overtones generated by this pipe which a normal person can hear is:

(A) 14 **(B)** 13 **(C)** 6 **(D)** 9

SPACE FOR ROUGH WORK

6. A coil has 1,000 turns and 500 cm^2 as its area. The plane of the coil is placed at right angles to a magnetic field of $2\times10^{-5}\,Wb/m^2$. The coil is rotated through 180° about its diameter in 0.2 *seconds*. The average e.m.f. induced in the coil, in *milli-volts*, is:

(A) 0 **(B)** 10 **(C)** 15 **(D)** 20

7. A conducting circular loop of radius r carries a constant current i. It is placed in a uniform magnetic field $\vec{B}$, such that $\vec{B}$ is perpendicular to the plane of the loop. The magnitude of magnetic force acting on the loop is:

(A) irB **(B)** $2\pi riB$ **(C)** Zero **(D)** πriB

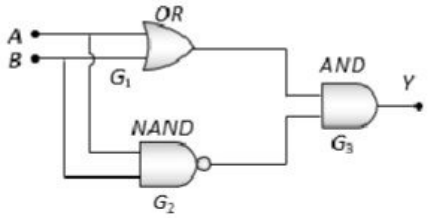
8. The work function of a metal is 4.2 *eV*, its threshold wavelength will be:

(A) 4000Å **(B)** 3500Å **(C)** 2952Å **(D)** 2500Å

9. The depth d at which the value of acceleration due to gravity becomes $\frac{1}{n}$ times the value at the surface, is [R = radius of the earth]:

(A) $\frac{R}{n}$ **(B)** $R\left(\frac{n-1}{n}\right)$ **(C)** $\frac{R}{n^2}$ **(D)** $R\left(\frac{n}{n+1}\right)$

10. Output Y for the following configuration of gate is:



(A) AB **(B)** $\bar{A}B+A\bar{B}$ **(C)** $A+B$ **(D)** $\bar{A}B$

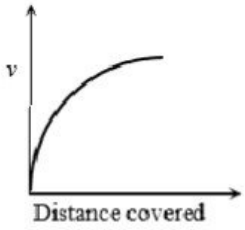
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11. A thin copper wire of length l *metre* increases in length by 2% when heated through 10°C. What is the percentage increase in area when a square copper sheet of side length l *metre* is heated through 10°*C*.

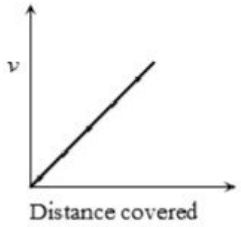
(A) 4% **(B)** 8% **(C)** 16% **(D)** 2%

12. A lead shot of 1*mm* diameter falls through a long column of glycerine. The variation of its velocity *v*. with distance covered is represented by:

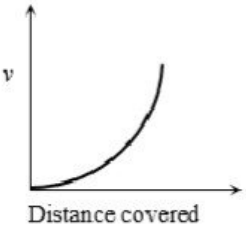
(A)



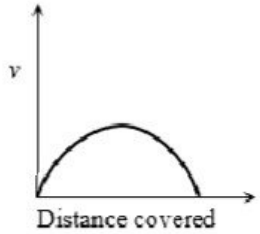
(B)



(C)



(D)



13. The maximum speed that can be achieved without skidding by a car on a circular unbanked road of radius R and coefficient of static friction μ, *is*:

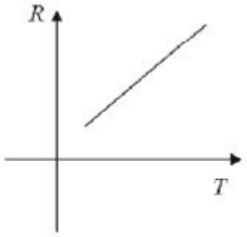
(A) $\frac{\sqrt{\mu Rg}}{2}$ **(B)** $\frac{\sqrt{Rg\mu}}{3}$ **(C)** $\frac{\sqrt{\mu Rg}}{4}$ **(D)** $\sqrt{\mu Rg}$

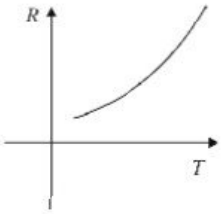
14. According to Newton, the viscous force acting between liquid layers of area A and velocity gradient $\Delta v/\Delta z$ is given by $F = -\eta A\frac{\Delta v}{\Delta z}$ where η is constant called coefficient of viscosity. The dimension of η are:

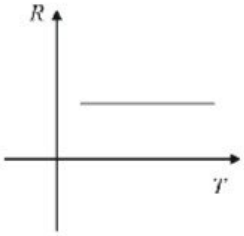
(A) $[ML^2T^{-2}]$ **(B)** $[ML^{-1}T^{-1}]$ **(C)** $[ML^{-2}T^{-2}]$ **(D)** $[M^0L^0T^0]$

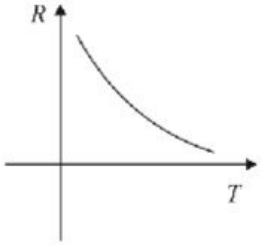
SPACE FOR ROUGH WORK

15. Graph which represents variation of resistance of semiconductor with temperature is:

(A) 

(B) 

(C) 

(D) 

16. Two electrons each are fixed at a distance '$2d$'. A third charge proton placed at the midpoint is displaced slightly by a distance $x\ (x << d)$ perpendicular to the line joining the two fixed charges. Proton will execute simple harmonic motion having angular frequency: (m = mass of charged particle)

(A) $\left(\frac{2q^2}{\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$ **(B)** $\left(\frac{\pi\varepsilon_0 md^3}{2q^2}\right)^{\frac{1}{2}}$ **(C)** $\left(\frac{q^2}{2\pi\varepsilon_0 md^3}\right)^{\frac{1}{2}}$ **(D)** $\left(\frac{2\pi\varepsilon_0 md^3}{q^2}\right)^{\frac{1}{2}}$

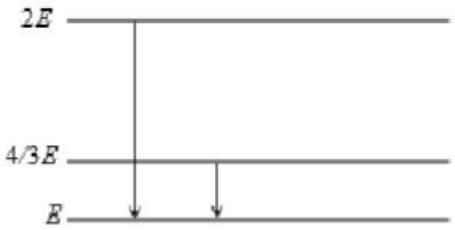
17. A ball is projected upwards from a height h above the surface of the earth with velocity v. The time at which the ball strikes the ground is:

(A) $\frac{v}{g}+\sqrt{\frac{2h}{g}}$ **(B)** $\frac{2v}{g}+\sqrt{\frac{2h}{g}}$

(C) $\frac{v}{g}\left[1+\sqrt{1+\frac{2gh}{v^2}}\right]$ **(D)** $\frac{v}{g}\left[1+\sqrt{1+\frac{gh}{v^2}}\right]$

SPACE FOR ROUGH WORK

18. The following diagram indicates the energy levels of a certain atom when the system moves from $2E$ level to E, a photon of wavelength λ is emitted. The wavelength of photon produced during its transition from $\frac{4E}{3}$ level to E is:



(A) $\lambda/3$ **(B)** $3\lambda/4$ **(C)** $4\lambda/3$ **(D)** 3λ

19. A body is sliding down an inclined plane having coefficient of friction 0.5. If magnitude of the normal reaction is twice that of the magnitude of resultant downward force along the incline, the angle between the inclined plane and the horizontal is:

(A) 15° **(B)** 30° **(C)** 45° **(D)** 60°

20. A bomb of mass 30 *kg* at rest explodes into two pieces of masses 18 *kg* and 12 *kg*. The velocity of 18 *kg* mass is $6\,ms^{-1}$. The kinetic energy of the other mass is:

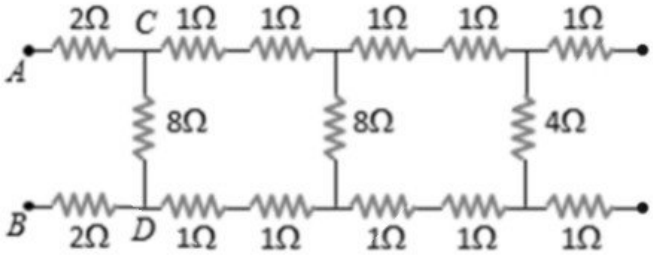
(A) $256\,J$ **(B)** $486\,J$ **(C)** $524\,J$ **(D)** $324\,J$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. In the figure shown, the equivalent resistance between A and B is _____ Ω.



2. Two coherent beams of light having intensities I and $4I$ interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\frac{\pi}{2}$ at point A and π at point B. Then the difference between the resultant intensities at A and B is xI, x is _____.

3. A uniform cube is subjected to volume compression. If each side is decreased by 1% , then bulk strain is $\frac{x}{100}$, x is _____.

4. A particle executing simple harmonic motion has an amplitude of 6 *cm*. Its acceleration at a distance of 2 *cm* from the mean position is $8 cm/s^2$. The maximum speed of the particle is _____ cm/s.

5. In a transformer, the coefficient of mutual inductance between the primary and the secondary coil is 0.2 *henry*. When the current changes by 5 *ampere/second* in the primary, the induced e.m.f. in the secondary will be _____ *V*.

SPACE FOR ROUGH WORK

6. A body of 2 kg mass makes an elastic collision with another body at rest. The velocity of the 2 kg mass is reduced to one-fourth of the original velocity. However, direction remains unchanged. The mass of the body struck is $M \times 10^{-1} kg$. Find the value of M.

7. The mass defect for the nucleus of helium is 0.0303 $a.m.u.$ The binding energy per nucleon for helium in MeV is _____. (Closest integral value)

8. The capacity of a parallel plate condenser is $5\mu F$. When a dielectric is placed between the plates of the condenser, (after removing the cell) potential difference across its plates becomes $1/8th$ of the original value. The value of dielectric constant will be _____.

9. A solid sphere rolls on a smooth horizontal surface at 10 m/s and then rolls up a smooth inclined plane of inclination 30° with horizontal. The mass of the sphere is 2 kg. Find the maximum height (in m) attained by the sphere. ($g = 10 m/s^2$)

10. A proton moving with a velocity of $2.5 \times 10^7 \, m/s$, enters a magnetic field of intensity 2.5T making an angle 30° with the magnetic field. The force on the proton is $n \times 10^{-12} N$, n is _____.

SPACE FOR ROUGH WORK

PART II: CHEMISTRY	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Consider the following statements:

According to Werner's theory,

(1) Ligands are connected to the metal ions by pure covalent bonds.

(2) Secondary valencies have directional properties.

(3) Secondary valencies are non-ionisable.

Of these statements:

(A) 1, 2 and 3 are correct **(B)** 2 and 3 are correct

(C) 1 and 3 are correct **(D)** 1 and 2 are correct

2. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A) : The ease of dehydration of alcohol follows the order : Primary < Secondary <Tertiary.

Reason (R) : Hydroxide ion is good leaving group.

In the light of the above statements, choose the correct answer from the options given below:

(A) A is false but R is true

(B) Both A and R are true but R is not the correct explanation of A

(C) Both A and R are true and R is the correct explanation of A

(D) A is true but R is false

3. Given below are two statements:

Statement-1: The IUPAC name of $CH_3CH = CH - C \equiv CH$ is pent-2-en-4-yne and not pent-3-en-1-yne.

Statement-2: Priority order according to IUPAC system of nomenclature $\rightarrow$ Substituent > Multiple bond > Functional group.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement-1 is incorrect, Statement-2 is correct

(B) Both Statement-1 and Statement-2 are correct

(C) Statement-1 is correct but Statement-2 is incorrect

(D) Both Statement-1 and Statement-2 are incorrect

SPACE FOR ROUGH WORK

4. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): pK_a value of ortho-nitrophenol is 7.2 while that of para-nitrophenol is 7.1.

Reason (R)s: ortho-nitrophenol is a stronger acid than para-nitrophenol.

In the light of the above statements, choose the correct answer from the options given below:

(A) A is false but R is true

(B) A is true but R is false

(C) Both A and R are true but R is not the correct explanation of A

(D) Both A and R are true and R is the correct explanation of A

5. Match List I with List II:

List I		List II (Deficiency Disease)	
A	Vitamin A	I	Rickets
B	Vitamin B	II	Beri beri
C	Vitamin C	III	Night blindness
D	Vitamin D	IV	Scurvy

Choose the correct answer from the options given below:

(A) A – IV, B – I, C – III, D – II **(B)** A – III, B – II, C – IV, D – I

(C) A – III, B – IV, C – II, D – I **(D)** A – IV, B – I, C – II, D – III

6. The mixture that shows negative deviation from Raoult's law is:

(A) Benzene + Toluene **(B)** Hexane + Heptane

(C) Water + Ethanol **(D)** Water + Nitric acid

7. For the given reaction, choose the correct expression of K_p from the following:

$$CaCO_{3(s)} \rightleftharpoons CaO_{(s)} + CO_{2(g)}$$

(A) $\frac{P_{CaCO_3}}{P_{CaO_{(s)}}.P_{CO_2}}$ **(B)** $\frac{1}{P_{CO_2}}$ **(C)** P_{CO_2} **(D)** $\frac{P_{CaO}.P_{CO_2}}{P_{CaCO_3}}$

8. The correct sequence of electronegativity of the elements listed below is:

F, O, C, H

(A) $F > O > C > H$ **(B)** $F > O > H > C$

(C) $O > H > C > F$ **(D)** $O > H > F > C$

SPACE FOR ROUGH WORK

9. The Molecular orbitals have specific characters:

A. Have the same energy

B. Have more than one nuclei

C. The molecular orbitals are filled in accordance with Aufbau principle

D. Types of molecular orbitals are bonding molecular orbitals and antibonding molecular orbitals

Choose the most appropriate from the options given below:

(A) B and D only **(B)** B, C, D only **(C)** A and C only **(D)** A, B, C only

10. The product **(C)** in the below mentioned reaction sequence is:

$$CH_3-CHBr-CH_2-Br \xrightarrow{Zn\,dust} A \xrightarrow[(C_6H_5CO)_2O_2]{HBr} B \xrightarrow{KCN} C$$

(A) 2- Methylpropane nitrile **(B)** Propane nitrile

(C) Butane nitrile **(D)** Ethane nitrile

11. The factor from the following that does not affect electronic conductance of a solution is:

(A) The nature and structure of metal **(B)** The number of valence electron per atom

(C) Temperature **(D)** Pressure

12. Match List I with List II:

List A (Technique)		List B (Application)	
A	Lassaigne's Test	**I**	Estimation of nitrogen in the form of Nitrogen gas
B	Duma's method	**II**	Detection of N, S & halogens
C	Kjeldahl's method	**III**	Estimation of halogen
D	Carius method	**IV**	Estimation of nitrogen in the form of ammonia gas

Choose the correct answer from the options given below:

(A) A- IV, B- I, C-II, D- III **(B)** A- I, B- II, C-III, D- IV

(C) A- II, B- I, C-IV, D- III **(D)** A- II, B- IV, C-I, D- III

SPACE FOR ROUGH WORK

13. Identify correct statements from below:

A. Bronze is an alloy of copper and tin.

B. Interstitial hydrides of *d*-block elements are non-stoichiometric.

C. Potassium dichromate is used as a primary standard in volumetric analysis.

D. $KMnO_4$ is used for the bleaching of wool, cotton, silk and other textile fibres.

E. Tetrahedral permanganate ion is blue in colour.

Choose the correct answer from the options given below:

(A) A, B, C only **(B)** A, B, E only **(C)** A, B, C, D only **(D)** B, C, E only

14. The compound that is Blood Red in colour is:

(A) $[Fe(CN)_5NOS]^{4-}$ **(B)** $[Fe(SCN)]^{2+}$

(C) PbS **(D)** $AgCl$

15. Given below are two statements:

Statement-I : Argon is used in the laboratory for handling substances that are air-sensitive.

Statement-II : Argon is inert towards chemical reactivity due to the completely filled valence shell electronic configuration, high ionization enthalpy and more positive electron gain enthalpy.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is true but Statement *II* is false

(B) Both Statement I and Statement II are true

(C) Both Statement *I* and Statement II are false

(D) Statement I is false but Statement II is true

16. Consider the oxides of group 13 elements:

B_2O_3, Al_2O_3, Ga_2O_3, Tl_2O_3, In_2O_3. The basic oxides are:

(A) B_2O_3, Al_2O_3 **(B)** Ga_2O_3, Tl_2O_3

(C) Al_2O_3, In_2O_3 **(D)** In_2O_3, Tl_2O_3

SPACE FOR ROUGH WORK

17. A species having positive charge but can not act as an electrophile is:

(A) $C_2H_5^+$ **(B)** NH_4^+ **(C)** Cl^+ **(D)** CH_3^+

18. Which of the following purification method is based on difference in the solubilities of the components of the mixture?

(A) Sublimation **(B)** Crystallisation

(C) Chromatography **(D)** Distillation

19. The methods that are used to prevent rusting of iron are:

A. Covering of metal surface with paint

B. Covering of metal surface with bisphenol

C. Covering of metal surface with Zn

D. Covering of metal surface with Sn

E. Covering of metal surface with Cu

Choose the correct answer from the options given below:

(A) A, B, C, D and E **(B)** A, B, C and E

(C) A, B, C and D **(D)** B, C, D and E

20. The expression for k in terms of P_0, P_t and n for the following first order gas phase reaction is given by:

$$A(g) \longrightarrow nB(g)$$

(where P_t is total pressure at time t and P_0 is initial pressure)

(A) $k = \frac{2.303}{t}\log\frac{P_0(n-1)}{nP_0 - P_t}$ **(B)** $k = \frac{2.303}{t}\log\frac{nP_0(n-1)}{nP_0 - P_t}$

(C) $k = \frac{2.303}{t}\log\frac{P_0}{nP_0 - P_t}$ **(D)** $k = \frac{2.303}{t}\log\frac{P_0(n-1)}{P_0 - P_t}$

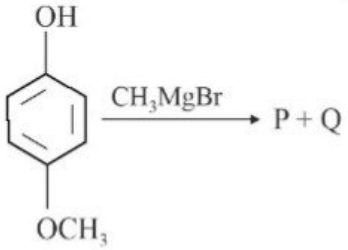
SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

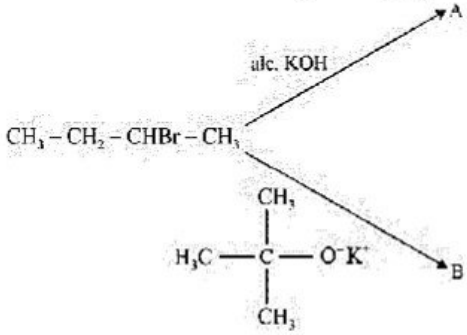
The answer to each question should be **rounded off to the nearest integer**.

1. The products of the following reaction are P and Q out of which, Q is a gas.



The molar mass of gas Q is _____ g/mol.

2. The sum of number of hyperconjugable hydrogens in the major products A and B is _____.



3. One of the salt from Na_2SO_3, Na_2S and Na_2CO_3 which evolves colourless gas with suffocating odour on heating with dil. H_2SO_4 has a molar mass of _____ g/mol.

(Molar mass in g/mol Na : 23, S : 32, C : 12, O : 16)

4. Number of moles of ethane required to produce 0.25 moles CO_2 after combustion is $x \times 10^{-3}$ moles. The value of x is _____.

5. The spin only magnetic moment for $\left[FeF_6\right]^{3-}$ is _____ $x10^{-2}$ BM. [Nearest integer]

SPACE FOR ROUGH WORK

6. Number of moles of gases obtained on electrolysis of one mole of sodium succinate is_____.

7. Two faraday of electricity liberates X grams of zinc from $ZnSO_4$. Value of X is _____.

$[M^{\circ}_{Zn} = 65\ g\ mol^{-1}]$

8. For the gaseous reaction $K + F \rightarrow K^+ + F^-, \Delta H$ was calculated to be 18.4 kcal/mol under conditions where the cations and anions were prevented from combining with each other. The ionization enthalpy of K is 4.3 eV/atom. If $X \times 10^{-1}$ eV is the magnitude of electron gain enthalpy of F, then X = _____.

(Take: 1 eV/atom = 23 kcal/mol)

9. The number of species from the following in which the central atom uses sp^2 hybrid orbitals in its bonding is _____.

$CO_2, NH_3, PCl_5, SO_2, SO_3, PCl_3, SF_6$

10. ΔG° for $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightleftharpoons NH_3$ is $-16.56\ kJ\ mol^{-1}$. K_p for the reaction at 27°C is _____.

(Nearest integer). [Given: ln10 = 2.3, R = 8J $mol^{-1}\ K^{-1}$]

SPACE FOR ROUGH WORK

PART III: MATHEMATICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Let $f(x)$ be a polynomial of least degree such that

$$\lim_{x\to 0}\left(1+\frac{x^2+f(x)}{x^2}\right)^{\frac{1}{x}}=e^2 \text{ and } h(x)=\begin{cases} f(x)+b & x\le 0 \\ \dfrac{e^x-e^{x\cos x}}{x+\sin x} & x>0 \end{cases} \text{ is continuous at } x=0 \text{ then } b \text{ is equal}$$

to:

(A) 0 **(B)** 1 **(C)** 4 **(D)** 5

2. An urn contains 4 green balls and 6 blue balls. A second urn contains 16 green balls and n blue balls. A single ball is drawn at random from each urn and probability that both balls are of same colour is $\frac{29}{50}$.

If first two drawn balls (one from each urn) are both green and probability of further two drawn balls (one from each urn and without replacing first two balls) are of same colour is p, then p is equal to:

(A) $\frac{5}{159}$ **(B)** $\frac{96}{159}$ **(C)** $\frac{101}{159}$ **(D)** $\frac{91}{159}$

3. The co-ordinates of the points A and B are respectively $(-3,2)$ and $(2, 3)$. P and Q are points on the line joining A and B such that $AP=PQ=QB$. A square $PQRS$ is constructed on PQ as one side, the co-ordinates of R can be:

(A) $\left(-\frac{4}{3},\frac{7}{3}\right)$ **(B)** $\left(\frac{2}{3},\frac{13}{3}\right)$ **(C)** $\left(\frac{1}{3},\frac{8}{3}\right)$ **(D)** $\left(\frac{2}{3},1\right)$

4. The number of integral value of m less than 10 for which the equation

$$\frac{\left(1+m^2\right)x^2-2(1+3m)x+(1+8m)}{2x^2+6x+9}>0 \ \forall x\in R:$$

(A) 9 **(B)** 3 **(C)** 0 **(D)** ∞

SPACE FOR ROUGH WORK

5. For $a>b$ and $c>\frac{b}{3}$, Let $(a-2b+3c)x^2+(a-c)x-2a+2b-2c=0$ and $\alpha\in(1,2)$ be one of its root then among two statements:

(i) $2a-3b+4c>0$

(ii) $a-b+c<0$

(A) Only (i) is true **(B)** Only (ii) is true

(C) Both (i) and (ii) are true **(D)** Neither (i) nor (ii) is true

6. If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$, then the value of $x^2+y^2+z^2+2xyz$ is:

(A) 1 **(B)** 0 **(C)** $x+y+z$ **(D)** $x^2y^2z^2$

7. General solution of the equation $(x\cos x)\frac{dy}{dx}+y(x\sin x+\cos x)=1$ is:

(A) $y\sec x=\tan x+c$ **(B)** $xy\sec x=\tan x+c$

(C) $xy=\sec x\tan x+c$ **(D)** $y\sec x=x\tan x+c$

8. Let $f(x)=Max\left\{x^2,(1-x)^2,2x(1-x)\right\}$ where $0\le x\le 1$. Then the area of the region bounded by the curve $y=f(x)$, the x-axis and the lines $x=0,\ x=1$ is:

(A) $\frac{17}{7}$ **(B)** $\frac{17}{27}$ **(C)** $\frac{14}{17}$ **(D)** $\frac{14}{27}$

SPACE FOR ROUGH WORK

9. If the sum of coefficient of all the positive even power of x in the binomial expansion of $\left(2x^3+\frac{3}{x}\right)^{10}$ is $\alpha^{10}-\beta.3^9$, and $cx^2+dx+e=0$ and $\beta x^2+\alpha x+4=0$ have a common root where $\alpha,\beta,c,d,e\in I^+$ then the sum of least possible value of $c+d+e$.

(A) 92 **(B)** 97 **(C)** 83 **(D)** 87

10. Let X denotes the sum of the numbers obtained when two fair dice are rolled. The variance of X are:

(A) $\sqrt{\frac{31}{6}}$ **(B)** $\frac{35}{6}$ **(C)** $\frac{17}{6}$ **(D)** $\frac{31}{6}$

11. If the circle $x^2+y^2+4x+22y+c=0$ bisects the circumference of the circle $x^2+y^2-2x+8y-d=0(c,d>0)$, then the maximum possible value of cd is:

(A) 25 **(B)** 125 **(C)** 425 **(D)** 625

12. If $\vec{a}.\vec{b}=\vec{a}.\vec{c}$ and $\vec{a}\times\vec{b}=\vec{c}\times\vec{a}$ ($\vec{a},\vec{b},\vec{c}\neq 0$ and $\vec{a},\vec{b},\vec{c}$ are different vectors) then which of the following can't be true?

(A) $|\vec{b}|=|\vec{c}|$

(B) $\vec{a}$ is parallel to a diagonal of a parallelogram with co-terminus edges as $\vec{b}$ and $\vec{c}$

(C) $\vec{a}$ is perpendicular to a diagonal of a parallelogram with co-terminus edges as $\vec{b}$ and $\vec{c}$

(D) $|\vec{a}|=|\vec{b}|+|\vec{c}|$

SPACE FOR ROUGH WORK

13. Find the sum to 10 terms of the series:

$$\frac{1}{1+1^2+1^4}+\frac{2}{1+2^2+2^4}+\frac{3}{1+3^2+3^4}+\ldots$$

(A) $\frac{45}{109}$ **(B)** $\frac{55}{109}$ **(C)** $\frac{45}{111}$ **(D)** $\frac{55}{111}$

14. If the solution of $\left(1+2e^{\frac{x}{y}}\right)dx+2e^{\frac{x}{y}}\left(1-\frac{x}{y}\right)dy=0$ is $\left(x+\lambda ye^{\frac{x}{y}}\right)=c$; ($c$ is an arbitrary constant), they λ is:

(A) 1 **(B)** 2 **(C)** 3 **(D)** 6

15. Let $f(x)=\frac{\tan^{-1}\left(1-\{x\}\right).\cos^{-1}\left(1-\{x\}\right).\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}}\left(1-\{x\}\right)^2}$

$\{x\}$ denotes fractional part of x, then $\lim\limits_{x\to 0^+} f(x)$:

(A) $\frac{\pi}{2\sqrt{2}}$ **(B)** $\frac{\pi^2}{8}$ **(C)** $\frac{\pi}{2}$ **(D)** 1

16. If the system of equations $ax+by+cz=0$; $bx+cy+az=0$; $cx+ay+bz=0$ has a non-trivial solution, then the system of equations.

$(b+c)x+(c+a)y+(a+b)z=0$;

$(c+a)x+(a+b)y+(b+c)z=0, (a+b)x+(b+c)y+(c+a)z=0$ has:

(A) Only one solution **(B)** No solution

(C) Infinite number of solutions **(D)** None of these

SPACE FOR ROUGH WORK

17. A function on $f:R\to R$ is such that $f\left(\frac{4x-8}{x-4}\right)=x$ for all $x\neq 4$ then the value of $fofof\left(\frac{1}{x}\right)$ at $x=\frac{1}{6}$, is:

(A) 10 **(B)** 8 **(C)** 6 **(D)** 12

18. It is given that

$$\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = xA+B$$

Where A and B are determinants of order 3 not involving x. Then $\det(A+B)$ is equal to:

(A) 12 **(B)** 24 **(C)** 15 **(D)** 18

19. Line L_1 is parallel to vector $\vec{a}=-3\hat{i}-2\hat{j}+4\hat{k}$ and passes through a point $A(7, 6, 2)$ and Line L_2 is parallel to vector $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}$ and passes through a point $B(5,3,4)$. Now a line L_3 parallel to vector $\vec{r}=2\hat{i}-2\hat{j}-\hat{k}$ intersect the line L_1 and L_2 at points C and D, respectively, then $|CD|=$

(A) 12 **(B)** 8 **(C)** 7 **(D)** 9

20. The points (5, 12) and (24, 7) are the foci of an ellipse passing through origin. The eccentricity of an hyperbola is $\sqrt{2}$ times the greatest integer less than reciprocal of eccentricity of given ellipse if the distance between foci of the hyperbola is 16, then equation of hyperbola is:

(A) $x^2-y^2=32$ **(B)** $x^2-y^2=16$

(C) $\frac{x^2}{16}-\frac{y^2}{9}=1$ **(D)** $\frac{x^2}{32}-\frac{y^2}{16}=1$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. If $\frac{P}{3}=\int_0^{\pi/2}\frac{dx}{\left(\sqrt{\cos x}+\sqrt{\sin x}\right)^4}$, then P is equal to_____.

2. If $\vec{a}$ and $\vec{b}$ are any two vectors of magnitudes 1 and 2, respectively, and $\left(1-3\vec{a}.\vec{b}\right)^2+\left|2\vec{a}+\vec{b}+3\left(\vec{a}\times\vec{b}\right)\right|^2=47$, and the angle between $\vec{a}$ and $\vec{b}$ is α then $\left|2\cos\alpha\right|$ is _____.

3. Let $f(x)=\int_0^x 3^t\left(3^t-4\right)(x-t)\,dt\,(x\geq 0)$. If $x=a$ is the point where $f(x)$ attains its local minimum value then find the value of 3^a_____.

4. The number of combination of the letters taken from the words: **EXAMINATION** taking 4 at a time is:

5. A hyperbola has its axes along the coordinate axes, latus rectum is 8 and conjugate axis is half of the distance between the foci, if 2λ is half of the conjugate axis, then the value of $\sqrt{3}\lambda$ is _____.

SPACE FOR ROUGH WORK

6. Let $f(x)=\dfrac{e^x}{1+e^x}$, $I_1=\int\limits_{f(-a)}^{f(a)} xg\left(x(1-x)\right)dx$, $I_2=\int\limits_{f(-a)}^{f(a)} g\left(x(1-x)\right)dx$, then I_2/I_1 is:

7. In the expansion $\dfrac{\left(x^5+1\right)\left(1+\dfrac{5}{x}+\dfrac{10}{x^2}+\dfrac{10}{x^3}+\dfrac{5}{x^4}+\dfrac{1}{x^5}\right)}{\left(x^4-x^3+x^2-x+1\right)}$ $x\neq 0$, the sum of the coefficient of x^{-3} and constant term is _____.

8. Let $A=\left\{x:\dfrac{x}{x-4}<0, x\in N\right\}$ and $R=\{(1,2),(2,3)\}$

Let S be the equivalence relation on A such that $R\subset S$ and the number of element in S is n, then minimum value of n.

9. If $Q(\alpha,\beta,\gamma)$ is the foot of the perpendicular from $P(2,4,3)$ on the line joining the points $A(1,2,4)$ and $B(3,4,5)$, then $9(\alpha+\beta+\gamma)$ _____.

10. If $\alpha+i\beta=\left(\dfrac{-1+i\sqrt{3}}{2}\right)^{\frac{3n_1}{4}}(1-i)^{-2n_2}$, (where n_1 and n_2 are positive integer). If P is the least value of n_1+n_2 for which α is 0 and Q is the least value of n_1+n_2 for which β is 0, then value of $P+Q$.

SPACE FOR ROUGH WORK

End of 2024 | Mock JEE Main - 9 (CBT)

Solutions to Mock JEE MAIN – 9 (CBT) | JEE 2024

PHYSICS

SECTION-1

1.(D) Given that,

$E_0 = 4V/m$

Average energy density of magnetic field,

$$= \frac{(B_{rms})^2}{2\mu_0} = \frac{\left(B_0/\sqrt{2}\right)^2}{2\mu_0} = \frac{1}{4}\frac{B_0^2}{\mu_0}$$

Average energy density of magnetic field is $\frac{1}{4}\frac{E_0^2}{c^2\mu_0}$

Average energy density of magnetic field,

$$= \frac{1}{4}\times 8.854\times 10^{-12}\times (4)^2 = 35.4\times 10^{-12} Jm^{-3}$$

2.(D) Average velocity becomes zero as motion of the molecule is random.

3.(B) $W_{AB} = \frac{nR\Delta T}{1-\gamma}$; $W_{BC} = \frac{nR\Delta T}{1-\gamma}$

Since, ΔT is same for AD and BC, $\therefore W_{AD} = W_{BC}$

4.(D) $\frac{\sin\frac{A+\delta m}{2}}{\sin\frac{A}{2}} = \mu$, But $\frac{A+\delta_m}{2} = i = 45^\circ$

So $\frac{\sin 45^\circ}{\sin(A/2)} = \sqrt{2} \Rightarrow \frac{1}{2} = \sin\frac{A}{2} \Rightarrow A = 60^\circ$

5.(C) Maximum hearing frequency for a person is 20,000*Hz*.

If a closed pipe vibrating in N^{th} mode then frequency of vibration $n = \frac{(2N-1)v}{4l} = (2N-1)n_1$

(where n_1 = fundamental frequency of vibration)

Hence $20{,}000 = (2N-1)\times 1500 \Rightarrow N = 7.1 \approx 7$

Also, in closed pipe

Number of over tones = (No. of mode of vibration) –1

= 7 – 1 = 6.

6.(B) By using $\varepsilon = -\frac{NBA(\cos\theta_2 - \cos\theta_1)}{\Delta t}$

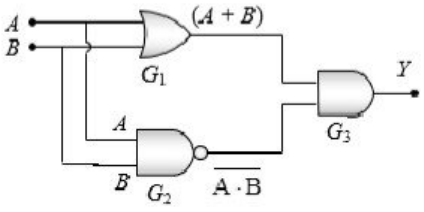
$$\varepsilon = -\frac{1000\times 2\times 10^{-5}\times 500\times 10^{-4}\left(\cos 180^o - \cos 0^o\right)}{0.2} = 10^{-2} volt = 10mV$$

7.(C) Net force on a current carrying closed loop is always zero, if it is placed in an uniform magnetic field.

8.(C) $W_0(eV) = \frac{12400}{\lambda_0} \Rightarrow \lambda_0 = \frac{12400}{4.2} \approx 2952\,\text{Å}$

9.(B) $g' = g\left(1-\frac{d}{R}\right) \Rightarrow \frac{g}{n} = g\left(1-\frac{d}{R}\right) \Rightarrow d = \left(\frac{n-1}{n}\right)R$

10.(B)



$Y = (A+B).\overline{AB}$

The given output equation can also be written as

$Y = (A+B).(\bar{A}+\bar{B}) = A\bar{A} + A\bar{B} + B\bar{A} + B\bar{B} = 0 + A\bar{B} + \bar{A}B + 0 = \bar{A}B + A\bar{B}$

11.(A) Since percentage increase in length = 2 %

Hence, percentage increase in area of square sheet

$= 2 \times 2\% = 4\%$

As $\beta = 2\alpha$

12.(A) Initially velocity increases due to effect of gravity. After some time it will attain terminal velocity.

13.(D) In the given condition the required centripetal force is provided by frictional force between the road and tyre.

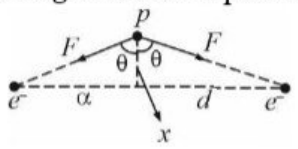
$$\frac{mv^2}{R} = \mu mg \quad \therefore \quad v = \sqrt{\mu Rg}$$

14.(B) $F = -\eta A\frac{\Delta v}{\Delta z} \Rightarrow [\eta] = [ML^{-1}T^{-1}]$

As $F = [MLT^{-2}]$, $A = [L^2]$, $\frac{\Delta v}{\Delta z} = [T^{-1}]$

15.(D)

16.(C) Net restoring force on the proton,



$F_{net} = 2F\cos\theta = \frac{2q^2}{4\pi\epsilon(d^2+x^2)} \cdot \frac{x}{(d^2+x^2)^{1/2}}$

As $x << d$, $F_{net} \approx \left[\frac{q^2}{2\pi\epsilon_0 d^3}\right] x$

Therefore, the angular frequency, $w = \left(\frac{q^2}{2\pi\epsilon_0 m d^3}\right)^{1/2}$

17.(C) Since direction of v is opposite to the direction of g and h so from equation of motion

$$h = -vt + \frac{1}{2}gt^2$$

$$\Rightarrow gt^2 - 2vt - 2h = 0 \Rightarrow t = \frac{2v \pm \sqrt{4v^2 + 8gh}}{2g} \Rightarrow t = \frac{v}{g}\left[1 + \sqrt{1 + \frac{2gh}{v^2}}\right]$$

18.(D) $2E - E = \frac{hc}{\lambda} \Rightarrow E = \frac{hc}{\lambda}$

$$\frac{4E}{3} - E = \frac{hc}{\lambda'} \Rightarrow \frac{E}{3} = \frac{hc}{\lambda'} \quad \therefore \frac{\lambda'}{\lambda} = 3 \Rightarrow \lambda' = 3\lambda$$

19.(C) Resultant downward force along the incline $= mg(\sin\theta - \mu\cos\theta)$

Normal reaction $= mg\cos\theta$

Given : $mg\cos\theta = 2mg(\sin\theta - \mu\cos\theta)$

By solving $\theta = 45^\circ$.

20.(B)



According to law of conservation of linear momentum

$m_A v_A = m_B v_B = 18\times 6 = 12\times v_B \Rightarrow v_B = 9\ m/s$

K.E. of mass 12 kg, $E_B = \frac{1}{2}m_B v_B^2 = \frac{1}{2}\times 12\times (9)^2 = 486J$

SECTION-2

1.(8) The last two resistance are out of circuit. Now 8Ω is in parallel with $(1+1+4+1+1)\Omega$.

$\therefore\ R_{CD} = 8\Omega\ ||8\Omega = \frac{8}{2} = 4\Omega\ \Rightarrow\ R_{AB} = 4+2+2 = 8\Omega$

2.(4) At point A, resultant intensity

$I_A = I_1 + I_2 = 5I$; and at point B

$I_B = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\pi = 5I - 4I$

$I_B = I$ so $I_A - I_B = 4I$.

3.(3) If side of the cube is L then $V = L^3 \Rightarrow \frac{dV}{V} = 3\frac{dL}{L}$

$\therefore$ % change in volume = 3 × (% change in length)

= 3 × 1% = 3% $\therefore$ Bulk strain $\frac{\Delta V}{V} = 0.03$

4.(12) $A = \omega^2 y\ \Rightarrow\ \omega = \sqrt{A/y} = \sqrt{\frac{8}{2}} = 2\ rad/\sec$

Now $v_{max} = A\omega = 6\times 2 = 12\ cm/s$

5.(1) $\varepsilon = M\frac{di}{dt} = 0.2\times 5 = 1V$

6.(12) Applying conservation of momentum, $2v + m\times 0 = 2\left(\frac{v}{4}\right) + mV$ or $mV = 2v - \frac{v}{2} = \frac{3v}{2}$ ...(i)

Since collision is elastic therefore kinetic energy is conserved

$\frac{1}{2}\times 2\times v^2 + \frac{1}{2}m\times 0\times 0 = \frac{1}{2}\times 2\left(\frac{v}{4}\right)^2 + \frac{1}{2}mV^2$ or $v^2 = \frac{v^2}{16} + \frac{1}{2}mV^2$ ...(ii)

Relative velocity before impact = $v - 0 = v$

Relative velocity after impact = $\frac{v}{4} - V$ Now, $v = -\left(\frac{v}{4} - V\right)$ or $v + \frac{v}{4} = V$ or $V = \frac{5v}{4}$

From eqn. (ii),

$$v^2=\frac{v^2}{16}+\frac{1}{2}m\frac{25v^2}{16} \quad \text{or} \quad 1=\frac{1}{16}+\frac{25m}{32} \quad or \quad \frac{15}{16}=\frac{25m}{32} \text{ or } \quad m=\frac{30}{25}=\frac{6}{5}=1.2kg$$

7.(7) $\frac{\text{Binding energy}}{\text{Nucleon}}=\frac{0.0303\times 931}{4}\approx 7$

8.(8) $V'=\frac{V}{8} \Rightarrow \frac{V}{K}=\frac{V}{8} \Rightarrow K=8$

9.(5) $\frac{1}{2}Mv^2+\frac{1}{2}I\omega^2=Mgh+\frac{1}{2}I\omega^2$ (as ω does not change)

$$\frac{1}{2}Mv^2=Mgh \Rightarrow h=\frac{v^2}{2g}=\frac{(10)^2}{2(10)}=5m$$

10.(5) $F=qvB\sin\theta=1.6\times10^{-19}\times2.5\times2.5\times10^7\sin30^o$

$$F=1.6\times10^{-19}\times6.25\times10^7\times\frac{1}{2}=5\times10^{-12}N$$

CHEMISTRY

SECTION-1

1.(B) Ligands are connected to the metal ion by coordinate bond or dative bond. Secondary valencies have directional properties. Secondary valencies are non-ionisable.

2.(D) The ease of dehydration of alcohol follows the order : Primary < Secondary <Tertiary. Reactivity of alcohol towards dehydration is based on stability of carbocation. –OH is a poor leaving group.

3.(D) The IUPAC name of $CH_3CH = CH - CCH$ is pent-3-en-1-yne, Priority order according to IUPAC → Functional group > Multiple bond > Substituent

4.(B) pK_a value is inversely proportional to the acidic strength. So para-nitrophenol is stronger acid than ortho-nitrophenol.

5.(B)

List I		List II	
A	Vitamin A	III	Night blindness
B	Vitamin B	II	Beri beri
C	Vitamin C	IV	Scurvy
D	Vitamin D	I	Rickets

6.(D) Benzene + Toluene ⇒ Ideal solution

Hexane + Heptane ⇒ Ideal solution

Water + Ethanol ⇒ Positive deviation from Raoult's law

Water + Nitric acid ⇒ Negative deviation from Raoult's law

7.(C) $K_p = \frac{P_{\text{gaseous product}}}{P_{\text{gaseous Reactant}}}$

Activity of Solid = 1, then $K_p = P_{CO_2}$

8.(A) $F = 4.0, O = 3.5, C = 2.5, H = 2.1$

9.(B) Different molecular orbitals have different energy, they have more than one nuclei. The molecular orbitals are filled in accordance with Aufbau principle and types of molecular orbitals are bonding molecular orbitals and antibonding molecular orbitals.

10.(C) $CH_3 - CHBr - CH_2 - Br \xrightarrow{Zn\,dust} A \xrightarrow[R_2O_2]{HBr} B \xrightarrow{KCN} C$

$A = CH_3 - CH = CH_2$

$B = CH_3 - CH_2 - CH_2 - Br$

$C = CH_3 - CH_2 - CH_2 - CN$ [Butane nitrile]

11.(D) Electronic conductance depends on the nature and structure of metal, the number of valence electron per atom and Temperature.

12.(C)

List A (Technique)		List B (Application)	
A	Lassaigne's test	**II**	Detection of N, S and halogens.
B	Duma's method	**I**	Estimation of nitrogen in form of Nitrogen gas
C	Kjeldahl's method	**IV**	Estimation of nitrogen in form of ammonia gas
D	Carius method	**III**	Estimation of halogen

13.(C) A. Bronze is an alloy of copper and tin.

B. Interstitial hydrides of d-block elements are non-stoichiometric.

C. Potassium dichromate is used as a primary standard in volumetric analysis.

D. $KMnO_4$ is used for the bleaching of wool, cotton, silk and other textile fibres.

E. Tetrahedral permanganate ion is purple in colour. (not blue)

14.(B) $[Fe(CN)_5NOS]^{4-}$ = Violet

$[Fe(SCN)]^{2+}$ = Blood red

PbS = Black

AgCl = white

15.(B) Both Statement I and Statement II are true. Argon is used in the laboratory for handling substances that are air-sensitive. Argon is inert towards chemical reactivity due to the completely filled valence shell electronic configuration, high ionization enthalpy and more positive electron gain enthalpy.

16.(D) B_2O_3 is Acidic ; Al_2O_3 and Ga_2O_3 are amphoteric; Tl_2O_3 and In_2O_3 are basic.

17.(B) In ammonium ion, the nitrogen atom has no vacant orbital in which it can accept an electron pair.

18.(B) Sublimation is used to separate sublimable compounds from non-sublimable impurities.

Crystallisation is based on 'Difference in the solubilities of the compounds' principle.

Chromatography is based on Adsorption or differential partitioning.

Distillation is used to separate Volatile liquids from non-volatile impurities and the liquids having sufficient difference in their boiling points.

19.(C) The methods that are used to prevent Rusting of iron are covering of metal surface with paint, covering of metal surface with bisphenol, covering of metal surface with Zn and covering of metal surface with Sn. Reactivity of copper is less than Fe, So it cannot be used to prevent rusting of iron.

20.(A) It is a 1st order reaction

Let initial pressure at time t is P_0.

$A(g)$	$\longrightarrow$	$nB(g)$	
P_0		0	$t=0$
(P_0-x)		nx	
$(=P_A)$			

$\therefore$ P_t (Total pressure at time 't') $= P_0 - x + nx = P_0 + (n-1)x$

$\therefore$ $x = \dfrac{P_t - P_0}{n-1}$

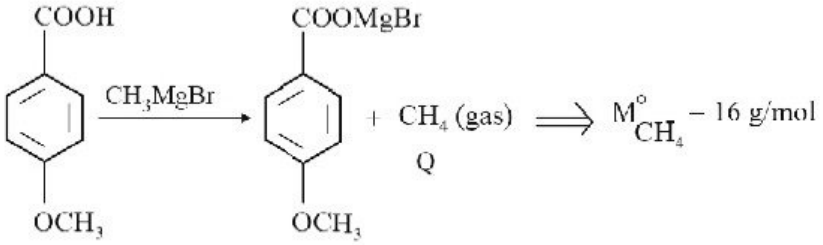
$\therefore$ $P_A = P_0 - \dfrac{P_t - P_0}{n-1} = \dfrac{P_0 n - P_t}{n-1}$

$\therefore$ $a \propto P_0 \;\&\; (a-x) \propto P_A = \dfrac{nP_0 - P_t}{n-1}$

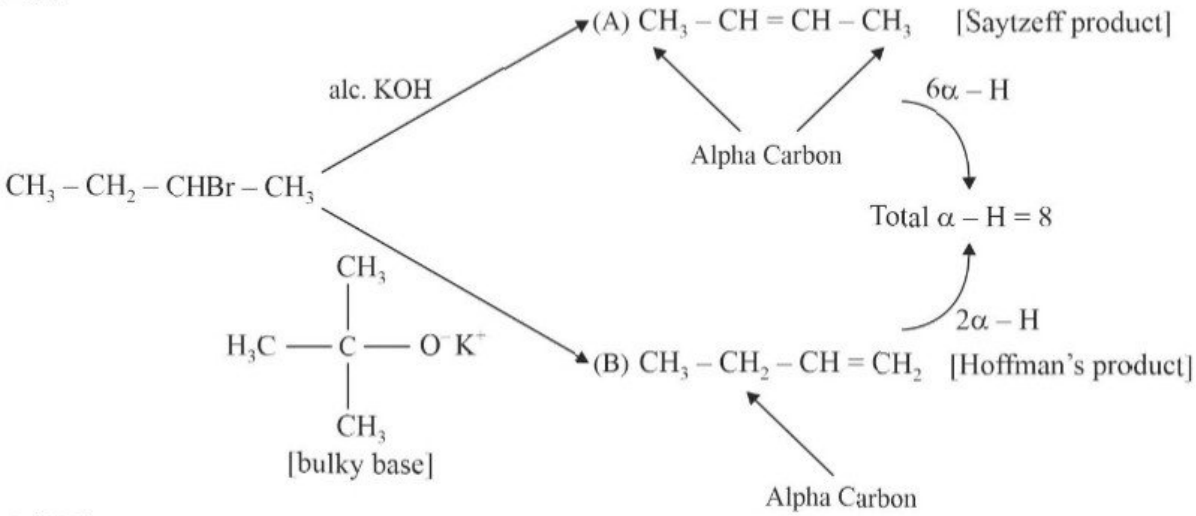
$\therefore$ $k = \dfrac{2.303}{t} \log \dfrac{P_0(n-1)}{nP_0 - P_t}$

SECTION-2

1.(16)



2.(8)



3.(126)

$Na_2CO_3 + dil.\ H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + CO_2 \uparrow$ (Colourless and odourless gas)

$Na_2SO_3 + dil.\ H_2SO_4 \longrightarrow Na_2SO_4 + H_2O + SO_2 \uparrow$ (Colourless and Suffocating odour gas)

$Na_2S + dil.\ H_2SO_4 \longrightarrow Na_2SO_4 + H_2S \uparrow$ (Colourless and pungent gas)

Na_2SO_3 evolves colourless gas with suffocating odour on heating with dil. H_2SO_4

Molar mass of Na_2SO_3 is 126 g/mol

4.(125)

$2C_2H_6 + 7O_2 \longrightarrow 4CO_2 + 6H_2O$

4 moles of CO_2 is produced from 2 moles of ethane

1 mole of CO_2 is produced from $\frac{2}{4}$ moles of ethane

0.25 moles of CO_2 is produced from $\frac{2}{4} \times 0.25 = 0.125$ moles of ethane

$= 125 \times 10^{-3}$ moles of ethane $\Rightarrow X = 125$

5.(592)

$[FeF_6]^{3-}$ Oxidation number of iron = +3

Then no. of unpaired electron (n)= 5 [F^- : WFL, $\therefore$, no pairing occurs]

$\mu = \sqrt{n(n+2)} = \sqrt{5(5+2)} = \sqrt{35} = 5.916 = 591.6 \times 10^{-2}$ BM

$\simeq 592 \times 10^{-2}$ BM

6.(4)

$$\begin{array}{l} CH_2COONa \\ | \\ CH_2COONa \end{array} + 2H_2O \xrightarrow{\text{electrolysis}} C_2H_4 \uparrow + 2CO_2 \uparrow + H_2 \uparrow + 2NaOH$$

1 mole ethene, 1 mole hydrogen gas, 2 moles of CO_2 $\Rightarrow$ total moles of gases = 4 moles

7.(65) Molar mass of zinc = 65 gm

n- factor = 2

$$E = \frac{M_0}{n} = \frac{65}{2} = 32.5\,gm,\ Q = 2F$$

$$W = ZQ = \frac{E}{F}Q = \frac{32.5}{F} \times 2F = 65gm \Rightarrow X = 65$$

8.(35) $K(g) + F(g) \rightarrow F^-(g) + K^+(g)$ $\quad \Delta H = 18.4\,kCal = 0.8\,eV$

$K(g) \rightarrow K^+(g) + e^-$ $\quad IE = 4.3\,eV$

$F(g) + e^- \rightarrow F^-(g)$

$$\Delta_{eg}H = \Delta H - IE = 0.8 - 4.3 = -3.5\ eV = -35 \times 10^{-1} eV$$

$$\Rightarrow |\Delta_{eg}H| = 35 \times 10^{-1} eV$$

$$\Rightarrow X = 35$$

9.(2) $CO_2 - sp$ hybridised

$SO_2, SO_3 - sp^2$ hybridised

$NH_3, PCl_3 - sp^3$ hybridised

$PCl_5 - sp^3d$ hybridised

$SF_6 - sp^3d^2$ hybridised

10.(1000)

$$\log K_P = -\frac{\Delta G^\circ}{2.3RT} = \frac{+16.56 \times 10^3}{2.3 \times 8 \times 300} = 3$$

$$K_P = 1000$$

MATHEMATICS

SECTION-1

1.(A) Consider $\lim\limits_{x\to 0}\left(1+\dfrac{x^2+f(x)}{x^2}\right)=L(say)$

It exists only when $\lim\limits_{x\to 0}\dfrac{x^2+f(x)}{x^2}=0$ (i.e. it converts into 1^{∞} form), so the least degree of $f(x)$ would be 2. i.e., $f(x)=a_2x^2+a_3x^3+.....$

$\because \quad \lim\limits_{x\to 0}\left(1+\dfrac{x^2+f(x)}{x^2}\right)^{1/x}=e^2$

$\Rightarrow \quad e^{\lim\limits_{x\to 0}\left(\frac{x^2+f(x)}{x^2}\right)\times\frac{1}{x}}=e^2 \quad \Rightarrow \quad e^{\lim\limits_{x\to 0}\frac{x^2+f(x)}{x^3}}=e^2$

$\Rightarrow \quad \lim\limits_{x\to 0}\dfrac{x^2+f(x)}{x^3}=2 \quad \Rightarrow \quad \lim\limits_{x\to 0}\dfrac{x^2+a_2x^2+a_3x^3+...}{x^3}=2$

$\Rightarrow \quad a_2=-1, a_3=2$ and a_4, a_5 are any arbitrary constants. Since we want polynomial of least degree. Hence $f(x)=-x^2+2x^3$

Now, $\lim\limits_{x\to 0}\dfrac{(e^x-1)-(e^{x\cos x}-1)}{x+\sin x}$

$$=\lim_{x\to 0}\left(\frac{\left(e^x-1\right)}{x\left(1+\frac{\sin x}{x}\right)}-\frac{\left(e^{x\cos x}-1\right)}{x\cos x\left(\sec x+\frac{\sin x}{x\cos x}\right)}\right)$$

$$=\lim_{x\to 0}\frac{\left(e^x-1\right)}{x\left(1+\frac{\sin x}{x}\right)}-\lim_{x\to 0}\frac{\left(e^{x\cos x}-1\right)}{x\cos x\left(\sec x+\frac{\sin x}{x\cos x}\right)}=\frac{1}{2}-\frac{1}{2}=0$$

2.(C) $\underset{I}{\boxed{\begin{matrix}4G\\6B\end{matrix}}} \quad \underset{II}{\boxed{\begin{matrix}16G\\nB\end{matrix}}}$

$\dfrac{29}{50}=\dfrac{4}{10}.\dfrac{16}{n+16}+\dfrac{6}{10}.\dfrac{n}{n+16}\Rightarrow n=144$ and $p=\left(\dfrac{3}{9}.\dfrac{15}{159}+\dfrac{6}{9}.\dfrac{144}{159}\right)=\dfrac{101}{159}$

3.(D) P, Q divides AB in the ratio of 1 : 2 and 2 : 1 respectively and hence the co-ordinates of P and Q are

$\left[\dfrac{1\times 2+2(-3)}{3}, \dfrac{1\times 3+2\times 2}{3}\right]$ and $\left[\dfrac{2\times 2+1(-3)}{3}, \dfrac{2\times 3+1\times 2}{3}\right]$

$\Rightarrow \quad P\left(-\dfrac{4}{3}, \dfrac{7}{3}\right)$ and $Q\left(\dfrac{1}{3}, \dfrac{8}{3}\right)$ then $PQ=\dfrac{\sqrt{26}}{3}$

Let the co-ordinates of R be (x, y)

Then, $QR = PQ \Rightarrow \left(x-\frac{1}{3}\right)^2+\left(y-\frac{8}{3}\right)^2=\frac{26}{9}$ ...(i)

And QR is perpendicular to PQ

$$\Rightarrow \quad \frac{y-\frac{8}{3}}{x-\frac{1}{3}}\times\frac{\frac{8}{3}-\frac{7}{3}}{\frac{1}{3}+\frac{4}{3}}=-1 \quad \Rightarrow \quad \left(y-\frac{8}{3}\right)=-5\left(x-\frac{1}{3}\right) \qquad \ldots(ii)$$

From equation, (i) and (ii), we get

$$26\left(x-\frac{1}{3}\right)^2=26/9 \Rightarrow x-\frac{1}{3}=\pm\frac{1}{3} \quad \Rightarrow \quad x=0 \text{ or } x=2/3$$

From equation (ii), when $x=0, y=13/3$ and when $x=2/3, y=1$

So, the required co-ordinates of R are $\left(0,\frac{13}{3}\right)$ or $\left(\frac{2}{3},1\right)$

4.(A) $D<0$

$$\Rightarrow \quad \left(-2(1+3m)\right)^2-4\left(1+m^2\right)(1+8m)<0$$

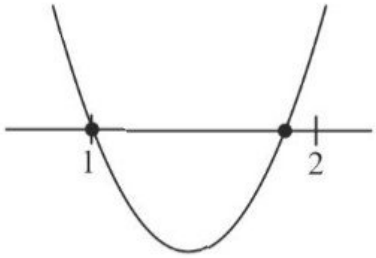
$$\Rightarrow \quad 4\left[(1+3m)^2-\left(1+m^2\right)(1+8m)\right]<0$$

$$\Rightarrow \quad \left[\left(1+9m^2+6m\right)-\left(1+m^2+8m+8m^3\right)\right]<0$$

$$\Rightarrow \quad -8m^3+8m^2-2m<0 \Rightarrow \quad (-2m)\left(4m^2-4m+1\right)<0$$

$$\Rightarrow \quad m(2m-1)^2>0 \quad \Rightarrow \quad m>0 \text{ and } m\neq\frac{1}{2} \quad \Rightarrow \quad m\in\left(0,\frac{1}{2}\right)\cup\left(\frac{1}{2},\infty\right)$$

5.(C)



$f(2)=4a-8b+12c+2a-2c-2a+2b-2c>0$

$4a-6b+8c>0$

$2a-3b+4c>0$, also $f(0)>0 \Rightarrow a-b+c<0$

6.(A) Let $\cos^{-1}x=\alpha, \cos^{-1}y=\beta$ and $\cos^{-1}z=\gamma$.

Therefore $\cos\alpha=x, \cos\beta=y$ and $\cos\gamma=z$ and at least one of x, y and z is non-negative. Now

$\alpha+\beta+\gamma=\pi \Rightarrow \alpha+\beta=\pi-\gamma$

Therefore $\cos(\alpha+\beta)=\cos(\pi-\gamma)=-\cos\gamma$

$\cos\alpha\cos\beta-\sin\alpha\sin\beta=-\cos\gamma$

This gives $xy-\sqrt{1-x^2}\sqrt{1-y^2}=-z$

$$(xy+z)^2=\left(1-x^2\right)\left(1-y^2\right)$$

$$x^2y^2+z^2+2xyz=1-x^2-y^2+x^2y^2$$

$$x^2+y^2+z^2+2xyz=1$$

7.(B) Given equation is:

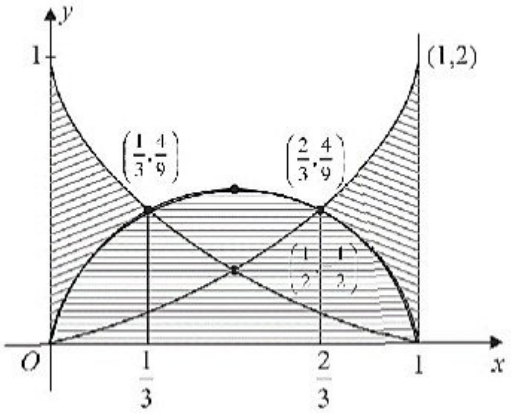
$$\frac{dy}{dx}+y\frac{(x\sin x+\cos x)}{x\cos x}=\frac{1}{x}\sec x$$

So $\frac{dy}{dx}+\left(\tan x+\frac{1}{x}\right)y=\frac{1}{x}\sec x$ (Linear)

The integrating factor is $I.F.=e^{\int\left(\tan x+\frac{1}{x}\right)dx}=e^{\log(x\sec x)}=x\sec x$

Hence the solution is $y(x\sec x)=\int\left(\frac{1}{x}\sec x\right)(x\sec x)\,dx+c=\tan x+c$

8.(B)



Let $C_1: y=x^2;\ C_2: y=(1-x)^2;\ C_3: y=2x(1-x)$. Then C_2 and C_3 intersect in $(1/3,4/9)$ and C_1 intersect C_3 in $(2/3,4/9)$.

$$\text{Now } f(x)=\begin{cases}(1-x)^2 & for \quad 0\le x\le 1/3\\ 2x(1-x) & for \quad 1/3\le x\le 2/3\\ x^2 & for \quad 2/3\le x\le 1\end{cases}$$

Required area (shaded portion)

$$=\int_0^{1/3}(1-x)^2\,dx+\int_{1/3}^{2/3}2x(1-x)\,dx+\int_{2/3}^{1}x^2dx=-\frac{1}{3}\left[(1-x)^3\right]_0^{1/3}+\left[x^2\right]_{1/3}^{2/3}-\frac{2}{3}\left[x^3\right]_{1/3}^{2/3}+\frac{1}{3}\left[x^3\right]_{2/3}^{1}$$

$$=-\frac{1}{3}\left[\frac{8}{27}-1\right]+\left(\frac{4}{9}-\frac{1}{9}\right)-\frac{2}{3}\left(\frac{8}{27}-\frac{1}{27}\right)+\frac{1}{3}\left(1-\frac{8}{27}\right)$$

$$=\frac{19}{81}+\frac{3}{9}-\frac{14}{81}+\frac{19}{81}=\frac{19+27-14+19}{81}=\frac{51}{81}=\frac{17}{27}$$

9.(A) $T_{r+1} = {}^{10}C_r \left(2x^3\right)^{10-r} \left(\frac{3}{x}\right)^r = {}^{10}C_r \, 2^{10-r} 3^r x^{30-4r}$

Put $r = 0,1,2,....7$ and we get sum of coefficient of all the positive power of x is equal to:

$5^{10} - 83 \times 3^9$

$\alpha = 5 \quad \beta = 83$

So $83x^2 + 5x + 4 = 0$ have both roots imaginary

So $\frac{c}{83} = \frac{d}{5} = \frac{e}{4} = 1$

$c + d + e = 92$

10.(B) Let X denotes the sum of the numbers obtained when two fair dice are rolled. So, X may have values, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 or 12.

(as 1 can't be the sum of two numbers on fair dice)

$P(X=2) = P\left[\{1,1\}\right] = \frac{1}{36}, P(X=3) = P\left[\{(1,2),(2,1)\}\right] = \frac{2}{36}$

$P(X=4) = P\left[\{(1,3),(2,2),(3,1)\}\right] = \frac{3}{36}$

$P(X=5) = P\left[\{(1,4),(2,3),(3,2),(4,1)\}\right] = \frac{4}{36}$

$P(X=6) = P\left[\{(1,5),(2,4),(3,3),(4,2),(5,1)\}\right] = \frac{5}{36}$

$P(X=7) = P\left[\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}\right] = \frac{6}{36}$

$P(X=8) = P\left[\{(2,6),(3,5),(4,4),(5,3),(6,2)\}\right] = \frac{5}{36}$

$P(X=9) = P\left[\{(3,6),(4,5),(5,4),(6,3)\}\right] = \frac{4}{36}$

$P(X=10) = P\left[\{(4,6),(5,5),(6,4)\}\right] = \frac{3}{36}$

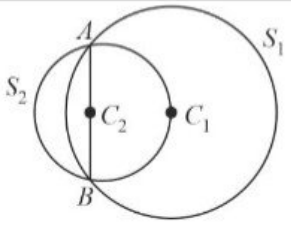
$P(X=11) = P\left[\{(5,6),(6,5)\}\right] = \frac{2}{36}$

$P(X=12) = P\left[\{(6,6)\}\right] = \frac{1}{36}$

$\text{Mean} = \sum XP(X) = \frac{252}{36} = 7$

$\text{Variance} = \sum X^2 P(X) - (Mean)^2 = \frac{35}{6}$

11.(D)



Given that $S_1 = x^2 + y^2 + 4x + 22y + c = 0$

$S_2 = x^2 + y^2 - 2x + 8y - d = 0$

AB is a common chord passing through C_2.

Equation of $AB: S_1 - S_2 = 0$

$\Rightarrow \quad \left(x^2 + y^2 + 4x + 22y + c\right) - \left(x^2 + y^2 - 2x + 8\right) = 0$

$\Rightarrow \quad 6x + 14y + c + d = 0$

The common chord passes through centre of S_2 i.e., $C_2(1, -4)$

$\Rightarrow \quad 6(1) + 14(-4) + c + d = 0 \quad \Rightarrow \quad c + d = 50$

Since $AM. \geq G.M.$

$\frac{c+d}{2} \geq \sqrt{cd} \quad \Rightarrow \quad 25 \geq \sqrt{cd} \quad \Rightarrow cd \leq 625$

12.(D) $\vec{a}.\vec{b} = \vec{a}.\vec{c}$

$\Rightarrow \quad \vec{a}.\left(\vec{b} - \vec{c}\right) = 0$

$\Rightarrow \quad \vec{a} \perp \left(\vec{b} - \vec{c}\right)$ ...(i)

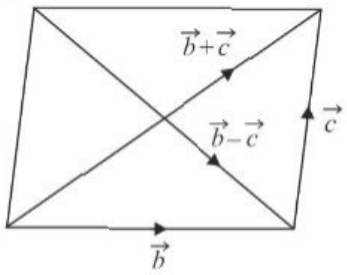
$\vec{a} \times \vec{b} = \vec{c} \times \vec{a}$

$\Rightarrow \quad \vec{a} \times \left(\vec{b} + \vec{c}\right) = \vec{0} \quad \Rightarrow \quad \vec{a} \parallel \left(\vec{b} + \vec{c}\right)$ ...(ii)

From (i) and (ii)

$\left(\vec{b} - \vec{c}\right).\left(\vec{b} + \vec{c}\right) = 0$

$\left|\vec{b}\right| = \left|\vec{c}\right|$



13.(D) $T_r = \frac{r}{1 + r^2 + r^4}$

$= \frac{r}{\left(1 + r^2 + r\right)\left(1 + r^2 - r\right)} = \frac{1}{2}\left[\frac{1}{1 + r^2 - r} - \frac{1}{1 + r^2 + r}\right] = \frac{1}{2}\left[V_{r-1} - V_r\right]$

So $T_1 = \frac{1}{2}\left[V_0 - V_1\right]$

$$T_2=\frac{1}{2}[V_1-V_2]$$

$$T_n=\frac{1}{2}[V_{n-1}-V_n]$$

$$\sum_{r=1}^{n}T_r=\frac{1}{2}[V_0-V_n]=\frac{1}{2}\left[1-\frac{1}{n^2+n+1}\right]=\frac{1}{2}\left[\frac{n^2+n}{n^2+n+1}\right]$$

14.(B) The appearance of $\frac{x}{y}$ in the given equation suggests the substitution $\frac{x}{y}=u$

$\Rightarrow dx=udy+ydu$

So, the given equation can be written as $\left(1+2e^u\right)(udy+ydu)+2e^u(1-u)dy=0$, which is a variable separable form

$$\Rightarrow \quad \frac{\left(1+2e^u\right)}{u+2e^u}du+\frac{dy}{y}=0 \qquad \Rightarrow \quad \int\left(\frac{1+2e^u}{u+2e^u}\right)du+\int\frac{dy}{y}=0$$

$$\Rightarrow \quad \log\left|u+2e^u\right|+\log|y|=\log|c| \quad \Rightarrow \quad \left(u+2e^u\right)y=c$$

$$\Rightarrow \quad \left(x+2ye^{x/y}\right)=c\text{; comparing with given general solution, we get } \lambda=2$$

15.(B) $$f(x)=\frac{\tan^{-1}\left(1-\{x\}\right)\cos^{-1}\left(1-\{x\}\right).\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2\{x\}}\left(1-\{x\}\right)^2}$$

$$\therefore \quad P=\lim_{x\to0^+}f(x)=\lim_{x\to0^+}\frac{\tan^{-1}(1-x)\cos^{-1}(1-x)\sin^{-1}(1-x)}{\sqrt{2x}(1-x)^2}$$

$$\left(\because x\to0^+\Rightarrow\{x\}=x\right)$$

$$=\lim_{x\to0^+}\frac{\tan^{-1}(1-x)}{(1-x)}.\frac{\sin^{-1}(1-x)}{(1-x)}\frac{\cos^{-1}(1-x)}{\sqrt{2x}}$$

$$=\frac{\pi}{4}.\frac{\pi}{2}\times\lim_{x\to0^+}\frac{\cos^{-1}(1-x)}{\sqrt{2}\sqrt{x}}=\frac{\pi^2}{8}\times\lim_{\theta\to0^+}\frac{\theta}{\sqrt{1-\cos\theta}}$$

$$\begin{pmatrix}\text{Assuming }\cos^{-1}(1-x)=\theta\\ \text{As } x\to0^+,1-x\in(0,1)\\ \Rightarrow\theta\in(0,\pi/2)\\ \Rightarrow\sin\theta>0\end{pmatrix}$$

$$=\frac{\pi^2}{8\sqrt{2}}\times\lim_{\theta\to0^+}\frac{\theta}{\sqrt{2\sin^2\theta/2}}=\frac{\pi^2}{8\sqrt{2}}\lim_{\theta\to0^+}\frac{\theta}{\sqrt{2}\left|\sin(\theta/2)\right|}$$

$$=\frac{\pi^2}{16}\lim_{\theta\to0^+}\frac{\theta}{\sin\theta/2} \qquad \left(\because\sin\left(\frac{\theta}{2}\right)>0\right)$$

$$=\frac{\pi^2}{16}\lim_{\theta\to0^+}2\left(\frac{\theta/2}{\sin\theta/2}\right)=\frac{\pi^2}{8}(1)=\frac{\pi^2}{8}$$

16.(C) For existence of a non-trivial solution of the first system $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$$\Rightarrow \quad (a+b+c)\begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix} = 0 \quad \Rightarrow \quad \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix} = 0 \text{ or } (a+b+c) = 0$$

$$\Rightarrow \quad \begin{vmatrix} 1 & b & c \\ 0 & c-b & a-c \\ 0 & a-b & b-c \end{vmatrix} = 0 \text{ or } (a+b+c) = 0$$

$$\Rightarrow \quad \begin{vmatrix} c-b & a-c \\ a-b & b-c \end{vmatrix} = 0 \text{ or } (a+b+c) = 0 \qquad \ldots(i)$$

$$\because \quad \Delta = \begin{vmatrix} b+c & c+a & a+b \\ c+a & a+b & b+c \\ a+b & b+c & c+a \end{vmatrix} = 0 \text{ for second system of equation.}$$

$$\text{Now, } \Delta = 2(a+b+c)\begin{vmatrix} 1 & c+a & a+b \\ 1 & a+b & b+c \\ 1 & b+c & c+a \end{vmatrix}$$

$$= 2(a+b+c)\begin{vmatrix} 1 & c+a & a+b \\ 0 & b-c & c-a \\ 0 & b-a & c-b \end{vmatrix} = 2(a+b+c)\begin{vmatrix} b-c & c-a \\ b-a & c-b \end{vmatrix}$$

$$= 2(a+b+c)\begin{vmatrix} c-b & a-c \\ a-b & b-c \end{vmatrix} = 0 \text{ (using equation (i))}$$

$\therefore$ The second system will have a non-trivial solution.

Note: Remember that the existence of one nontrivial solution implies existence of infinite number of solutions for homogeneous equations.

17.(B) If $f\left(\frac{4x-8}{x-4}\right) = x$ for all $x \neq 4$. Let $y = \frac{4x-8}{x-4} \Rightarrow x = \frac{4y-8}{y-4} \Rightarrow f(y) = \frac{4y-8}{y-4}$

$$\Rightarrow \quad f(f(x)) = x \text{ (from given)}$$

$$f(1/x) = \frac{4(1/x)-8}{(1/x)-4} = \frac{4-8x}{1-4x} \text{ for } x \neq 0, \frac{1}{4}$$

$$\therefore \quad f(f(1/x)) = f\left(\frac{4-8x}{1--4x}\right) = \frac{4\left(\frac{4-8x}{1-4x}\right)-8}{\left(\frac{4-8x}{1-4x}\right)-4} = \frac{\left(\frac{4-8x}{1-4x}\right)-2}{\left(\frac{1-2x}{1-4x}\right)-1} = \frac{4-8x-2+8x}{1-2x-1+4x} = \frac{2}{2x} = \frac{1}{x}$$

$$\Rightarrow \quad f\left(f\left(f\left(\frac{1}{x}\right)\right)\right) = f\left(\frac{1}{x}\right)$$

$$f\left(\frac{1}{x}\right)=f(6)=\frac{4\times6-8}{6-4}=\frac{16}{2}=8$$

18.(A) Let

$$\Delta=\begin{vmatrix}x^2+x & x+1 & x-2\\ 2x^2+3x-1 & 3x & 3x-3\\ x^2+2x+3 & 2x-1 & 2x-1\end{vmatrix}$$

$$\Delta=\begin{vmatrix}x^2+x & x+1 & x-2\\ -4 & 0 & 0\\ x^2+2x+3 & 2x-1 & 2x-1\end{vmatrix}\left[by\,R_2\to R_2-(R_1+R_3)\right]$$

$$\Delta=\begin{vmatrix}x & x+1 & x-2\\ -4 & 0 & 0\\ 2x+3 & 2x-1 & 2x-1\end{vmatrix}\left[by\,R_1\to R_1+\left(\frac{x^2}{4}\right)R_2\,and\,R_3\to R_3+\frac{x^2}{4}R_2\right]$$

$$\Delta=\begin{vmatrix}x & x+1 & x-2\\ -4 & 0 & 0\\ 3 & -3 & 3\end{vmatrix}(by\,R_3\to R_3-2R_1)=x\begin{vmatrix}1 & 1 & 1\\ -4 & 0 & 0\\ 3 & -3 & 3\end{vmatrix}+\begin{vmatrix}0 & 1 & -2\\ -4 & 0 & 0\\ 3 & -3 & 3\end{vmatrix}=xA+B$$

19.(D) Line $L_1=\vec{r}=7\hat{i}+6\hat{j}+2\hat{k}+\lambda\left(-3\hat{i}+2\hat{j}+4\hat{k}\right)$ and line L_2 is

$L_2:\ \vec{r}=5\hat{i}+3\hat{j}+4\hat{k}+\mu\left(2\hat{i}+\hat{j}+3\hat{k}\right)$

As point C lies on L_1 and point D lies on L_2

$$\Rightarrow\quad \overrightarrow{CD}=L_2-L_1=2\hat{i}+3\hat{j}-2\hat{k}+\lambda\left(-3\hat{i}+2\hat{j}+4\hat{k}\right)-\mu\left(2\hat{i}+\hat{j}+3\hat{k}\right)\qquad\ldots(i)$$

Since it is parallel to $2\hat{i}-2\hat{j}-\hat{k}$

So, direction ratios will be proportional:

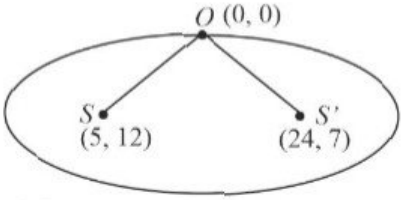
$$\Rightarrow\quad \frac{2-3\lambda-2\mu}{2}=\frac{3+2\lambda-\mu}{-2}=\frac{-2+4\lambda-3\mu}{-1}$$

On solving these equations, we get $\lambda=2,\mu=1$

Putting these values in (i),

$$\Rightarrow\quad \overrightarrow{CD}=-6\hat{i}+6\hat{j}+3\hat{k}\qquad\Rightarrow\quad\left|\overrightarrow{CD}\right|=9$$

20.(A)



$OS+OS'=38$

$SS'=38\times e$

$e=\frac{\sqrt{386}}{38}$

Eccentricity of hyperbola is $=\sqrt{2}\left[\frac{38}{\sqrt{386}}\right]=\sqrt{2}[1.9314]=\sqrt{2}$

Hence hyperbola is rectangular hyperbola

So Equation of hyperbola is $x^2-y^2=32$

SECTION-2

1.(1) $\int_0^{\pi/2}\frac{dx}{\cos^2 x\left(1+\sqrt{\tan x}\right)^4}=\int_0^{\pi/2}\frac{\sec^2 x}{\left(1+\sqrt{\tan x}\right)^4}dx$

Put $\tan x=t^2\Rightarrow \sec^2 x\,dx=2t\,dt$

$$\int_0^{\infty}\frac{2t}{(1+t)^4}dt=2\int_0^{\infty}\left(\frac{1}{(1+t)^3}-\frac{1}{(1+t)^4}\right)dt$$

$$=2\left(\frac{-1}{2(1+t)^2}+\frac{1}{3(1+t)^3}\right)_0^{\infty}=2\left(0-\left(\frac{-1}{2}+\frac{1}{3}\right)\right)=\frac{1}{3}$$

2.(1) $1+9\left(\vec{a}.\vec{b}\right)^2-6\left(\vec{a}.\vec{b}\right)+4|\vec{a}|^2+|\vec{b}|^2+9|\vec{a}\times\vec{b}|^2+4\vec{a}.\vec{b}=47$

Or $1+4+4+36-4\cos\alpha=47$

Or $\cos\alpha=-\frac{1}{2}$

Hence, the angle between $\vec{a}$ and $\vec{b}$ is $\frac{2\pi}{3}$.

3.(7) Given, $f(x)=x\int_0^x 3^t\left(3^t-4\right)dt-\int_0^x 3^t.t\left(3^t-4\right)dt$

For maxima/minima, we have

$$f'(x)=\int_0^x 3^t\left(3^t-4\right)dt$$

$$f'(x)=\frac{1}{2\ln 3}\left(3^{2x}-8.3^x+7\right)$$

$$f'(x)=\frac{1}{2\ln 3}\left(3^x-1\right)\left(3^x-7\right)$$

$f'(x)=0\Rightarrow x=0,\log_3 7$

$\therefore$ $x=\log_3 7$ is the point of minima.

Hence, $3^a=3^{\log_3 7}=7$

4.(136)

The word EXAMINATION contains; $AA, II, NN, E, X, M, T, O$

Then number of combinations = coefficient of x^4 in $\left(1+x+x^2\right)(1+x)^5$

$(\because 2A's, 2Is, 2N's, E, 1X, 1M, 1T \& 1O)$

= coefficient of x^4 in $\left\{(1+x)^3+x^6+3(1+x)^2x^2+3(1+x)x^4\right\}(1+x)^5$

= coefficient of x^4 in $\left\{(1+x)^8+x^6(1+x)^5+3x^2(1+x)^7+3x^4(1+x)^5\right\}$

$= {}^8C_4+0+3.{}^7C_2+3=\frac{8.7.6.5}{1.2.3.4}+3.\frac{7.6}{1.2}+3=70+63+3=136$

Aliter:

Case I : Four distinct letters

No. of ways = ${}^8C_4$

Case II : Two same and Two distinct letters

No. of ways = ${}^3C_1\times{}^7C_2$

Case III : Two same of one kind and two same of other kind of letters

No. of ways = ${}^3C_2$

$\therefore$ Total number of ways = ${}^8C_4+{}^3C_1\times{}^7C_2+{}^3C_2=136$

5.(6) Hyperbola is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

$b=\frac{ae}{2}$

$b^2=\frac{a^2e^2}{4}$

$a^2(e^2-1)=\frac{a^2e^2}{4}$

$e=\frac{2}{\sqrt{3}}$

$\frac{2b^2}{a}=8$

$\frac{2}{a}a^2(e^2-1)=8$

$a\left(e^2-1\right)=4$

$a=12$

6.(2) We have $f(a)+f(-a)=\frac{e^a}{1+e^a}+\frac{e^{-a}}{1+e^{-a}}$

$=\frac{e^a}{1+e^a}+\frac{1}{1+e^a}=1$

Therefore $I_1=\int\limits_{f(-a)}^{f(a)}(1-x)g\left((1-x)x\right)dx$

$=\int\limits_{f(-a)}^{f(a)}g\left(x(1-x)\right)dx-I_1=I_2-I_1$

So, $2I_1 = I_2 \Rightarrow \dfrac{I_2}{I_1} = 2$

7.(21) $\dfrac{\left(x^5+1\right)(1+x)^5}{x^5\left(x^4-x^3+x^2-x+1\right)} = \dfrac{(1+x)^6}{x^5}$

Constant term = coefficient of x^5 in $(1+x)^6 = {}^6C_5 = 6$

Coefficient of x^{-3} = coefficient of x^2 in $(1+x)^6 = {}^6C_2 = \dfrac{6\times5}{2} = 15$

8.(7)



$x \in (0,4)\ \&\ x \in N$

$A = \{1,2,3\}$

$R = \{(1,2),(2,3)\}$

So make equivalence relation on A for S.

Element added will be:

(1, 1) (2, 2) (3, 3) (2, 1) (3, 2) (3, 1) (1, 3)

9.(88) Let PQ be the perpendicular drawn from point $P(2,4,3)$ to the line joining the point $A(1,2,4)$ and $B(3,4,5)$

Let Q divides AB internally in the ratio $\lambda:1$

$\therefore \quad Q \equiv \left(\dfrac{3\lambda+1}{\lambda+1}, \dfrac{4\lambda+2}{\lambda+1}, \dfrac{5\lambda+4}{\lambda+1}\right)$ …(i)

Direction ratios of PQ are

$\dfrac{3\lambda+1}{\lambda+1}-2, \dfrac{4\lambda+2}{\lambda+1}-4, \dfrac{5\lambda+4}{\lambda+1}-3$

i.e., $\dfrac{\lambda-1}{\lambda+1}, \dfrac{-2}{\lambda+1}, \dfrac{2\lambda+1}{\lambda+1}$

Now, direction ratios of AB are, $3-1, 4-2, 5-4$ i.e., $2, 2, 1$.

Since PQ is perpendicular to AB.

$2\left(\dfrac{\lambda-1}{\lambda+1}\right) + \dfrac{2(-2)}{\lambda+1} + 1\left(\dfrac{2\lambda+1}{\lambda+1}\right) = 0$

$\therefore \quad \dfrac{2\lambda-2-4+2\lambda+1}{\lambda+1} = 0$

$\therefore \quad 4\lambda - 5 = 0$

$\therefore \quad 4\lambda = 5$

$\therefore \quad \lambda = \dfrac{5}{4}$

Putting $\lambda = \dfrac{5}{4}$ in (i),

Coordinates of Q are,

$$\left(\frac{3\left(\frac{5}{4}\right)+1}{\left(\frac{5}{4}\right)+1}=\frac{19}{19},\frac{4\left(\frac{5}{4}\right)+2}{\left(\frac{5}{4}\right)+1}=\frac{28}{9},\frac{5\left(\frac{5}{4}\right)+4}{\left(\frac{5}{4}\right)+1}=\frac{41}{9}\right)$$

$$\therefore \quad Q\equiv\left(\frac{19}{9},\frac{28}{9},\frac{41}{9}\right)$$

10.(5) $\alpha+i\beta=\left(\frac{-1+i\sqrt{3}}{2}\right)^{3n_1/4}(1-i)^{-2n_2}$

$\frac{-1+i\sqrt{3}}{2}=e^{i2\pi/3}$ and $1-i=\sqrt{2}e^{-i\pi/4}$

Now, $\alpha+i\beta=e^{i\frac{\pi n_1}{2}}\left(\sqrt{2}\right)^{-2n_2}.e^{i\frac{\pi}{2}.n_2}$

$\alpha+i\beta=2^{-n_2}e^{i(n_1+n_2)\pi/2}$

$\Rightarrow \quad \alpha=2^{-n_2}.\cos\left(n_1+n_2\right)\frac{\pi}{2}=0$

Where n_1+n_2 odd

$\beta=2^{-n_2}\sin\left(n_1+n_2\right)\frac{\pi}{2}=0$

Where n_1+n_2 is even


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